2 Fig. 2.1 shows a golfer about to hit a golf ball with a golf club. The initial momentum of the golf ball is zero.

Fig. 2.1

(a) Define momentum.

..... [1]

(b) The golf club is in contact with the ball for 5.0×10^{-4} s. The velocity of the golf ball as it leaves the golf club is 41 m / s. The golf ball has a mass of 0.046 kg.

(i) Calculate the impulse on the golf ball.

impulse = ... [2]

(ii) Calculate the force applied to the ball by the golf club.

force = ... [2]

[Total: 5]

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